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Lic. in Informatics and Computing Engineering (L.EIC)

Discrete Mathematics

3 - First Order Logic

Some of the following exercises are adapted from exercises in *"Logic in Computer Science: Modelling and Reasoning about Systems"*, by Michael Huth, and Mark Ryan.

3.1. Let $\mathcal{L}$ be a first order logic language, such that $\mathcal{F}_0 = \{\text{Ed}, \text{Patsy}, \text{Carl}, \text{Monique}\}$ and $\mathcal{R}_2 = \{F, M, H, S, B\}$. Let $F(x, y)$ mean that x is the father of y; $M(x, y)$ denotes x is the mother of y. Similarly, $H(x, y)$, $S(x, y)$ and $B(x, y)$ will represent the relations husband/sister/brother of, respectively. Translate the following sentences into first order logic:

(a) Everybody has a mother.

(b) Everybody has a father and a mother.

(c) Whoever has a mother has a father.

(d) Ed is a grandfather.

(e) Nobody's grandmother is anybody's father.

(f) Ed and Patsy are husband and wife.

(g) Carl is Monique's brother-in-law.

3.2. Translate each of the following propositions by a sentence in Portuguese or English; consider the set $\mathbb{R}$ of real numbers as the universe.

(a) $\exists x \, \forall y \, (x + y = y)$

(b) $\forall x \, \forall y \, (((x \geq 0) \wedge (y < 0)) \rightarrow (x - y > 0))$

(c) $\exists x \, \exists y \, (((x \leq 0) \wedge (y \leq 0)) \wedge (x - y > 0))$

(d) $\forall x \, \forall y \, (((x \neq 0) \wedge (y \neq 0)) \rightarrow (x \times y \neq 0))$

3.3. Which of the following strings are formulas in first order logic? Specify a reason for failure for strings that are not.

a) Let $\mathcal{F}_0 = \{m\}$, $\mathcal{F}_1 = \{f\}$ and $\mathcal{R}_2 = \{S, B\}$:

(a) $S(m, x)$

(b) $B(m, f(m))$

(c) $f(m)$

(d) $B(B(m, x), y)$

(e) $S(B(m), z)$
(f) $(B(x, y) \to (\exists z\, S(z, y)))$
(g) $(S(x, y) \to S(y, f(f(x))))$
(h) $(B(x) \to B(B(x)))$

b) Let $\mathcal{F}_0 = \{c, d\}$, $\mathcal{F}_1 = \{f\}$, $\mathcal{F}_2 = \{g\}$, $\mathcal{F}_3 = \{h\}$ and $\mathcal{R}_3 = \{P, Q\}$:
(a) $\forall x\, P(f(d), h(g(c, x), d, y))$
(b) $\forall x\, P(f(d), h(P(x, y), d, y))$
(c) $\forall x\, Q(g(h(x, f(d), x), g(x, x)), h(x, x, x), c)$
(d) $\exists z\, (Q(z, z, z) \to P(z))$
(e) $\forall x \forall y\, (g(x, y) \to P(x, y, x))$
(f) $Q(c, d, c)$

3.4. Let $\mathcal{L}$ be a first order logic language such that $\mathcal{F}_1 = \{f\}$, $\mathcal{F}_2 = \{g\}$ and $\mathcal{R}_2 = \{P, Q\}$. Let φ be $\exists x\, (Q(x, y) \wedge (\forall y \neg Q(y, x) \vee \exists z, P(y, z))))$.

a) Identify the free and bound occurrences of variables in φ.

b) Is there a variable in φ that has both free and bound occurrences?

c) Consider the terms w, $f(x)$ and $g(y, z)$:
(a) Compute $\varphi[x \mapsto w]$, $\varphi[y \mapsto w]$, $\varphi[y \mapsto f(x)]$ and $\varphi[z \mapsto g(y, z)]$.
(b) Which of w, $f(x)$ and $g(y, z)$ are free for x in φ?
(c) Which of w, $f(x)$ and $g(y, z)$ are free for y in φ?